

Paper 2 - Set 03 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	A	Binomial expansion
2	C	Sequences and series
3	B	Trigonometric functions and equations
4	A	Identifying errors in proofs
5	A	The logic of arguments
6	A	Identifying errors in proofs
7	A	The logic of arguments
8	A	Identifying errors in proofs
9	B	Exponentials and logarithms
10	C	Probability
11	H	Quadratic functions
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15	H	Number, units and estimation
16	D	Mathematical proof
17	E	GCSE algebra
18	F	Graphs of functions
19	F	Mathematical proof
20	F	Chains of reasoning

Worked solutions

Question 1 BIN-05 · Binomial expansion · spec MM2.4

Evaluate $\binom{10}{3} - \binom{10}{7} + \binom{9}{2}$.

Answer: A

$\binom{10}{3} = \binom{10}{7}$, so the first two cancel; $\binom{9}{2} = 36$.

Question 2 SEQ-02 · Sequences and series · spec MM2.3

Find the sum to infinity of the geometric series $12 - 4 + \frac{4}{3} - \frac{4}{9} + \dots$

Answer: C

$$a = 12, r = -\frac{1}{3}. S_{\infty} = \frac{12}{1 + \frac{1}{3}} = \frac{12}{4/3} = 9.$$

Question 3 TRG-09 · Trigonometric functions and equations · spec MM4.5

Simplify $\frac{1 - \cos^2 \theta}{\cos \theta \tan \theta}$ for values of θ where the expression is defined.

Answer: B

$\cos \theta \tan \theta = \sin \theta$, and $1 - \cos^2 \theta = \sin^2 \theta$. So the expression is $\sin \theta$.

Question 4 ERR-10 · Identifying errors in proofs · spec Err2

Claim: if $a > b$ then $\frac{1}{a} < \frac{1}{b}$.

I Suppose $a > b$.

II Dividing both sides by ab : $\frac{1}{b} > \frac{1}{a}$.

III So $\frac{1}{a} < \frac{1}{b}$.

Which of the following is correct?

Answer: A

$a = 1, b = -1$: $a > b$ but $\frac{1}{a} = 1 > -1 = \frac{1}{b}$. Dividing by ab without knowing its sign is invalid (and $b = 0$ would be division by zero).

Question 5 LGC-19 · The logic of arguments · spec Arg2

Let f be a function defined for all real numbers. P : ' $f'(a) = 0$ '; Q : ' f has a stationary point at $x = a$ '. Which one of the following is true?

Answer: A

A stationary point is by definition a point where the derivative is zero, so P and Q are equivalent.

Question 6 ERR-16 · Identifying errors in proofs · spec Err2

A student argues: 'Every prime number is odd, because if it were even it would be divisible by 2 and so not prime.' Which of the following is correct?

Answer: A

A number divisible by 2 is not prime unless it equals 2. The conclusion 'every prime is odd' is false.

Question 7 X2-24 · The logic of arguments · spec Arg2

For a real number x , P : ' $x > 0$ '. Q : ' $x^3 > 0$ '. Which one of the following is true?

Answer: A

Cubing preserves sign, so $x > 0 \Leftrightarrow x^3 > 0$.

Question 8 ERR-01 · Identifying errors in proofs · spec Err2

A student attempts to solve the equation $\sqrt{x+3} = x-3$ for real x .

I Squaring both sides: $x+3 = (x-3)^2 = x^2 - 6x + 9$.

II So $x^2 - 7x + 6 = 0$.

III So $(x-1)(x-6) = 0$.

IV Hence the solutions are $x = 1$ and $x = 6$.

Which of the following best describes the solution?

Answer: A

Squaring can introduce extraneous solutions, so line IV should check them: $x = 1$ gives $\sqrt{4} = 2 \neq -2$. Only $x = 6$ works. Lines I-III are valid deductions.

Question 9 X2-22 · Exponentials and logarithms · spec MM5.3

Consider the statement: **for all real x , $2^x > x^2$.** Which of the following is a counterexample?

Answer: B

$2^3 = 8 < 9 = 3^2$, so $x = 3$ breaks the claim. The other options satisfy it: $2 > 1$, $32 > 25$, $1 > 0$, $1024 > 100$.

Question 10 X2-19 · Probability · spec M7.7

Events A and B have $P(A) > 0$ and $P(B) > 0$. P : ' A and B are mutually exclusive'. Q : ' A and B are not independent'. Which one of the following is true?

Answer: C

Mutually exclusive gives $P(A \text{ and } B) = 0 \neq P(A)P(B)$, so they cannot be independent: P is sufficient. But two overlapping events can also fail to be independent, so P is not necessary.

Question 11 X2-01 · Quadratic functions · spec MM1.3

p and q are real numbers. Which of the following statements is/are true?

I If $p^2 < 4q$ then $x^2 + px + q > 0$ for all real x .

II If $x^2 + px + q > 0$ for all real x then $q > 0$.

III If $q < 0$ then $x^2 + px + q = 0$ has two real roots of opposite sign.

Answer: H

I: negative discriminant and positive leading coefficient. II: put $x = 0$. III: the product of the roots is $q < 0$ (and the discriminant $p^2 - 4q > 0$), so the roots are real with opposite signs.

Question 12 X2-11 · Functions and transformations · spec MM8.2

Let $f(x) = x^2$ for all real x , and let g be the function whose graph is obtained by translating the graph of $y = f(x)$ by 2 units in the positive x -direction and then reflecting in the y -axis. Which of the following statements is/are true?

- I $g(x) = (x + 2)^2$
- II $g(x) = (x - 2)^2$
- III The graph of g has its vertex at $(-2, 0)$

Answer: F

Translation gives $(x - 2)^2$; reflecting in the y -axis replaces x by $-x$: $(-x - 2)^2 = (x + 2)^2$, with vertex at $(-2, 0)$.

Question 13 RSN-13 · Chains of reasoning · spec Prf5

p and q are prime numbers with $p + q = 45$. Find pq .

Answer: D

An odd sum of two primes needs one of them to be 2 (the only even prime): $q = 43$, which is prime. $pq = 86$.

Question 14 RSN-11 · Chains of reasoning · spec Prf5

The product of three consecutive positive integers is N , and the middle one of the three integers is m . Which of the following statements must be true?

- I N is divisible by 6.
- II N is divisible by 4.
- III $N + m$ is a perfect cube.

Answer: F

Among three consecutive integers there is a multiple of 2 and a multiple of 3 (I). II fails for $1 \cdot 2 \cdot 3 = 6$. III: $N + m = (m - 1)m(m + 1) + m = m^3 - m + m = m^3$.

Question 15 X2-18 · Number, units and estimation · spec M2.3

n is a positive integer greater than 1. Which of the following statements is/are true?

- I If n is not prime, then n has a prime factor at most $\sqrt{n}$.
- II If n has no prime factor at most $\sqrt{n}$, then n is prime.
- III If n is prime, then n has no factor between 1 and n exclusive.

Answer: H

If $n = ab$ with $1 < a \leq b$ then $a \leq \sqrt{n}$ and a has a prime factor $\leq \sqrt{n}$ (I). II is the contrapositive of I. III is the definition of prime.

Question 16 PRF-05 · Mathematical proof · spec Prf3

The sums 1, $1 + 3$, $1 + 3 + 5$, $1 + 3 + 5 + 7$ are 1, 4, 9, 16. A student conjectures that the sum of the first n odd numbers is n^2 . Which of the following is a valid justification?

Answer: D

A applies the arithmetic-series formula to all n . (C also gives a correct calculation but B, D and E are not justifications.)

Question 17 ALG-08 · GCSE algebra · spec M4.19

The n th term of a sequence is $\frac{n^2 + 1}{2n}$. Which of the following statements is/are true?

- I The 5th term is 2.6.
- II Every term after the first is greater than the previous term.
- III Some term of the sequence equals 10.

Answer: E

I: $\frac{26}{10} = 2.6$. II: $u_n = \frac{n}{2} + \frac{1}{2n}$; $u_1 = 1$, $u_2 = 1.25$, and for $n \geq 1$ the increase $\frac{1}{2} - \frac{1}{2n(n+1)} > 0$, so increasing. III: $n^2 - 20n + 1 = 0$ has no integer solutions (discriminant 396 is not a square), so false.

Question 18 GRA-15 · Graphs of functions · spec MM8.6

The cubic $y = x^3 + ax + b$ has three distinct real roots. Which of the following statements must be true?

- I $a < 0$
- II $b \neq 0$
- III The curve has two turning points.

Answer: F

Three distinct real roots require two turning points (III), which needs $y' = 3x^2 + a$ to have two roots, so $a < 0$ (I). II is false: $x^3 - x$ has roots 0, ± 1 and $b = 0$.

Question 19 PRF-15 · Mathematical proof · spec Prf3

Let $u_n = n^2 - n + 41$. Which of the following statements is/are true?

- I u_n is prime for $n = 1, 2, 3, 4, 5$.
- II u_n is prime for all positive integers n .
- III u_{41} is divisible by 41.

Answer: F

$u_1 = 41$, $u_2 = 43$, $u_3 = 47$, $u_4 = 53$, $u_5 = 61$: all prime (I). $u_{41} = 41^2 - 41 + 41 = 41^2$, divisible by 41 (III) and not prime, so II is false - small cases do not prove a general claim.

Question 20 RSN-18 · Chains of reasoning · spec Prf5

How many ordered pairs of positive integers (m, n) satisfy $\frac{1}{m} + \frac{1}{n} = \frac{1}{4}$?

Answer: F

$4n + 4m = mn \Rightarrow (m - 4)(n - 4) = 16$. Positive factor pairs of 16 as ordered pairs: (1, 16), (2, 8), (4, 4), (8, 2), (16, 1): five. (Negative factor pairs would make m or n non-positive.)